

Assembly Language

Homework 1

Deadline: Wednesday October 8th, 2025 23:59

Number Systems

$$\begin{array}{r} 2 \overline{) 25} \\ \underline{20} \\ 5 \\ \underline{4} \\ 1 \end{array} \quad \begin{array}{r} 2 \overline{) 58} \\ \underline{20} \\ 38 \\ \underline{36} \\ 2 \\ \underline{2} \\ 0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 100} \\ \underline{20} \\ 80 \\ \underline{80} \\ 0 \end{array}$$

1. Convert the following decimal numbers to binary:

a. $25 = 11001_{(2)}$
b. $58 = 111010_{(2)}$
c. $100 = 1100100_{(2)}$

2. Convert the following binary numbers to decimal:

a. $1101 = 13$
b. $10110 = 22$
c. $111001 = 57$

3. Convert the following hexadecimal numbers to binary:

a. $1A = 00011010_{(2)}$
b. $3F = 00111111_{(2)}$
c. $7B = 0111011_{(2)}$

4. Convert the following binary numbers to hexadecimal:

a. $101011 = 2B$
b. $1101101 = 6D$
c. $11110000 = F0$

5. Perform binary addition for the following pairs of binary numbers:

a. $(1011 + 1101) = 11000$
b. $(100110 + 10101) = 111011$
c. $(11101 + 11011) = 111000$

$$\begin{array}{r} 100110 \\ + 10101 \\ \hline 111011 \end{array}$$

$$\begin{array}{r} 11101 \\ + 11011 \\ \hline 111000 \end{array}$$

A B C D E F
10 11 12 13 14 15

Data Representation

$$100 = 01100100_{(2)}$$

$$\begin{array}{r} 1's \quad 10011011 \\ + \quad \quad \quad 1 \\ \hline 10011100 \quad (2's) \end{array}$$

6. Represent the following signed decimal numbers in 8-bit two's complement form:

a. -12 = 11110100 (2's) 12 = 00001100₂ $\begin{array}{r} 2 \overline{)12} \\ 2 \overline{)6} - 0 \\ 2 \overline{)3} - 0 \\ 1 - 1 \end{array}$ $\begin{array}{r} 2 \overline{)45} \\ 2 \overline{)22} - 1 \\ 2 \overline{)11} - 0 \\ 2 \overline{)5} - 1 \\ 2 \overline{)2} - 1 \\ 1 - 0 \end{array}$

b. 45 = 00101101 (2's) 2's = 11101100

c. -100 = 10011100 (2's)

7. Represent the following unsigned decimal numbers in 8-bit binary form:

a. 200 = 11001000₂ $\begin{array}{r} 2 \overline{)200} \\ 2 \overline{)100} - 0 \\ 2 \overline{)50} - 0 \\ 2 \overline{)25} - 0 \\ 2 \overline{)12} - 1 \\ 2 \overline{)6} - 0 \\ 2 \overline{)3} - 1 \\ 1 - 0 \end{array}$ $\begin{array}{r} 2 \overline{)150} \\ 2 \overline{)75} - 0 \\ 2 \overline{)37} - 1 \\ 2 \overline{)18} - 1 \\ 2 \overline{)9} - 0 \\ 2 \overline{)4} - 1 \\ 2 \overline{)2} - 0 \end{array}$ $\begin{array}{r} 2 \overline{)200} \\ 2 \overline{)100} - 0 \\ 2 \overline{)50} - 0 \\ 2 \overline{)25} - 0 \\ 2 \overline{)12} - 1 \\ 2 \overline{)6} - 0 \\ 2 \overline{)3} - 1 \\ 1 - 1 \end{array}$ $\begin{array}{r} 2 \overline{)75} \\ 2 \overline{)37} - 1 \\ 2 \overline{)18} - 1 \\ 2 \overline{)9} - 0 \\ 2 \overline{)4} - 1 \\ 2 \overline{)2} - 0 \\ 1 - 0 \end{array}$

c. 75 = 01001011₂

8. Convert the following ASCII characters to their binary representation:

a. 'A' 65_{(10)}} = 1000001₍₂₎ $\begin{array}{r} 2 \overline{)65} \\ 2 \overline{)32} - 1 \\ 2 \overline{)16} - 0 \\ 2 \overline{)8} - 0 \\ 2 \overline{)4} - 0 \\ 2 \overline{)2} - 0 \\ 1 - 0 \end{array}$ $\begin{array}{r} 2 \overline{)122} \\ 2 \overline{)61} - 0 \\ 2 \overline{)30} - 1 \\ 2 \overline{)15} - 0 \\ 2 \overline{)7} - 1 \\ 2 \overline{)3} - 1 \\ 1 - 1 \end{array}$ $\begin{array}{r} 2 \overline{)53} \\ 2 \overline{)26} - 1 \\ 2 \overline{)13} - 0 \\ 2 \overline{)6} - 1 \\ 2 \overline{)3} - 0 \\ 1 - 1 \end{array}$

b. 'z' 122_{(10)}} = 1111010₍₂₎

c. '5' 53_{(10)}} = 110101₍₂₎

Boolean Expressions

9. Simplify the following Boolean expressions using Boolean algebra:

a. $(A \cdot \neg A + B \cdot \neg B) = 0 + 0 = 0$

b. $(A + \neg A \cdot B) = (A + \neg A)(A + B) = 1 \cdot (A + B) = A + B$

c. $((A + B) \cdot (\neg A + B)) = A \cdot \neg A + AB + \neg A \cdot B + B$
 $= 0 + B(A + \neg A) + B = B$

A	B	$\neg A$	$\neg A \cdot B$	$A + \neg A \cdot B$
0	0	1	0	0
0	1	1	1	1
1	0	0	0	1
1	1	0	0	1

10. Construct the truth table for the following Boolean expressions:

a.

A	B	C	$\neg C$	$B \cdot \neg C$	$A + B \cdot \neg C$
0	0	0	1	0	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	0	0	0
1	0	0	1	0	1
1	0	1	0	0	1
1	1	0	1	1	1
1	1	1	0	0	1

b. $\neg A \cdot B + C$

A	$\neg A$	B	C	$\neg A \cdot B$	$\neg A \cdot B + C$
0	1	0	0	0	0
0	1	0	1	0	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	0	1	1	0	1

c. $A \cdot B + \neg A \cdot \neg B$

A	B	$\neg A$	$\neg B$	$A \cdot B$	$\neg A \cdot \neg B$	$A \cdot B + \neg A \cdot \neg B$
0	0	1	1	0	1	1
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	1	0	0	1	0	1